MICROSCOPY

Microbiology is concerned with organisms so small they cannot be seen distinctly with the unaided eye. Because of this, a microscope is of crucial importance. Thus it is important to understand how different microscopes work for examination of image so that we can get a clear and distinct image.

Microscopy is used in many different applications:

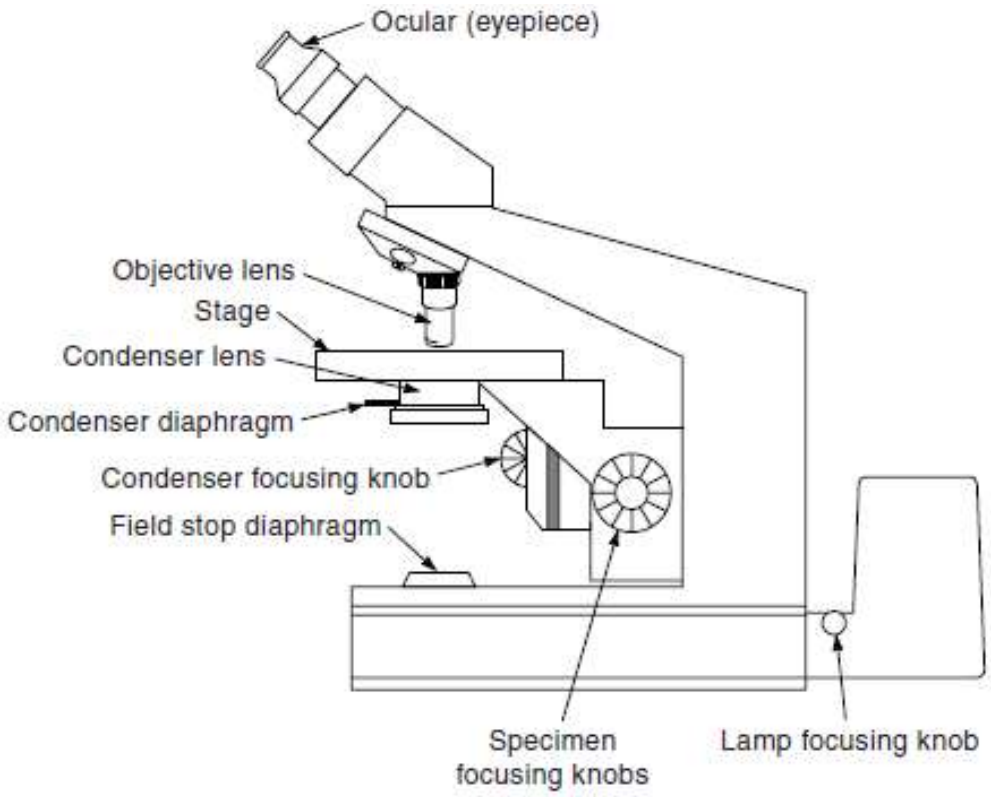
- to produce an **enlarged** image of an object.
- to evaluate the **integrity** of samples during an experiment,
- to map the fine **details** of the spatial distribution of macromolecules within cells
- to directly measure biochemical events within living tissues.

Microbiologists currently employ a variety of light microscopes in their work, eg.- bright-field, dark-field, phase-contrast etc., that are discussed below.

LIGHT MICROSCOPE

The simplest form of light microscope consists of a single glass lens mounted in a metal frame – a magnifying glass.

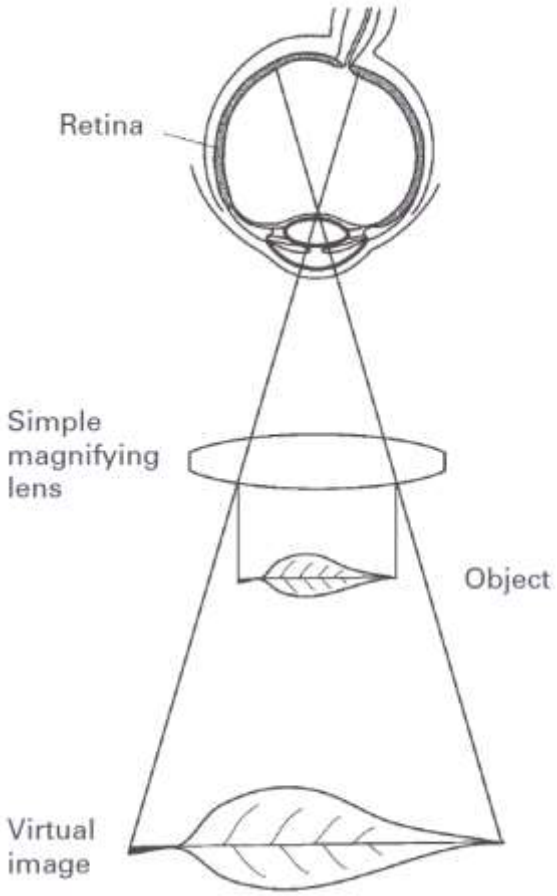
- Here the specimen requires very little preparation, and is usually held close to the eye in the hand.
- **Focusing** of the region of interest is achieved by moving the lens and the specimen relative to one another.
- The **source** of light is usually the sun or ambient indoor light.
- The **detector** is the human eye.



Light Microscope

MAGNIFICATION

- The magnification or linear magnification of a microscope is defined as the ratio of the image size to the object (specimen) size.
- If the image and object are in the same medium, then it is just the image distance divided by the object distance.
- There is a difference in the meaning of the two terms, magnification and **magnifying power**. Magnifying power or **angular magnification** is the ratio of the angle subtended by object and image.



FACTORS AFFECTING MAGNIFICATION

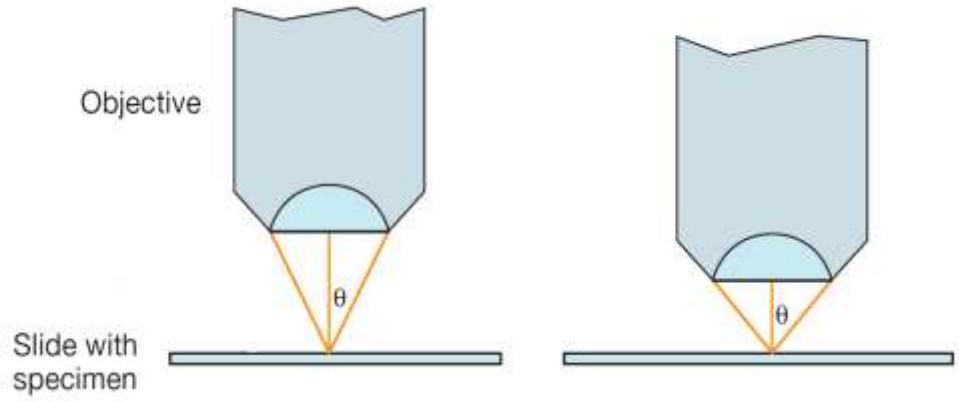
RESOLUTION AND NUMERICAL APERTURE

- The most important part of the microscope is the objective, which must produce a clear and magnified image
- Resolution is the ability of a lens to separate or distinguish between small objects that are close together.
- Much of the optical theory underlying microscope design was developed by the German physicist **Ernst Abbé,** in the 1870s.
- Resolution is described mathematically by an equation developed by Ernst Abbé, called **Abbé equation**

- The Abbé equation states that the minimal distance (d) between two objects that allows them to be seen as two separate entities, depends on the wavelength of light (λ) used to illuminate the specimen and on the numerical aperture of the lens (n sin θ), which is the ability of the lens to gather light.

  $d = 0.5 \lambda / n \sin \theta$

- As d becomes smaller, the resolution increases, and fine details can be observed in a specimen
- As d becomes smaller as the wavelength of light used decreases and as the numerical aperture (NA) increases.
- Thus the greatest resolution is obtained using a lens with the largest possible NA and light of the shortest wavelength
- The numerical aperture is the light gathering ability of a lens and is defined by two components: n is the refractive index of the medium in which the lens works (e.g., air) and θ is 1/2 the angle of the cone of light entering an objective.
- When this cone has a narrow angle and tapers to a sharp point, it does not spread out much after leaving the slide and therefore does not properly separate images of closely packed objects.
- If the cone of light has a very wide angle and spreads out rapidly after passing through a specimen, closely packed objects appear widely separated and are resolved.
- The angle of the cone of light that can enter a lens depends on the refractive index (n) of the medium in which the lens works, as well as upon the objective itself.
- The refractive index for air is 1.00 and sinθ cannot be greater than 1 (the maximum θ is 90° and sin 90° is 1.00).
- Therefore, no lens working in air can have a numerical aperture greater than 1.00.



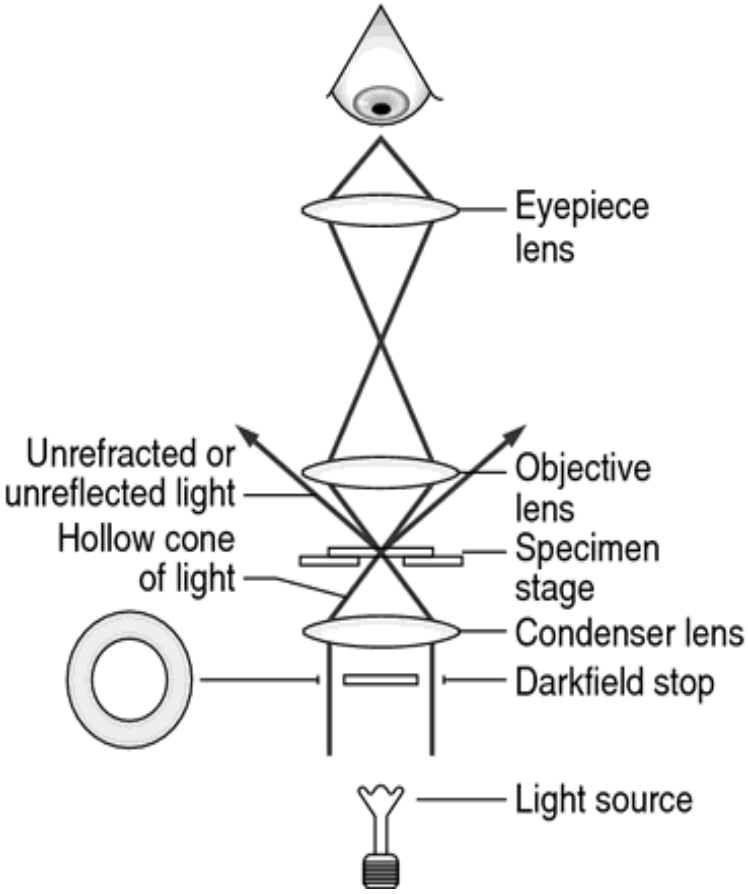
Numerical Aperture in Microscopy. The angular aperture θ is 1/2 the angle of the cone of light that enters a lens from a specimen, and the numerical aperture is n sinθ. In the right-hand illustration the lens has larger numerical apertures, resulting in greater resolution.

Important notes: These principles of microscopy; resolution, numerical aperture are important. Not only their definitions but in terms of numerical questions that can be asked from this section

BRIGHT FIELD MICROSCOPY

- Microscope is called a bright-field microscope because it forms a **dark image** against a **brighter background.**
- The microscope consists of a sturdy metal body or stand composed of a base and an arm to which the remaining parts are attached.
- A **light source**, either a mirror or an electric illuminator, is located at the base.
- Two focusing knobs, the **fine and coarse adjustment knobs**, are located on the arm and can move either the stage or the nosepiece to focus the image.
- The stage is positioned about halfway up the arm and holds microscope slides by either simple slide clips or a mechanical stage clip.
- A mechanical stage allows the operator to move a slide around smoothly during viewing by use of stage control knobs.
- The substage condenser is mounted within or beneath the stage and focuses a cone of light on the slide.
- Its position often is fixed in simpler microscopes but can be adjusted vertically in more advanced models.
- The curved upper part of the arm holds the ordinary body assembly, to which a nosepiece and one or more eyepieces or oculars are attached.

- More advanced microscopes have eyepieces for both eyes and are called **binocular microscopes.**
- The body assembly itself contains a series of mirrors and prisms so that the barrel holding the eyepiece may be tilted for ease in viewing.
- It is opposite to Dark field microscopy in which a brighter image is formed against a darker background.



A bright-field microscope

Important notes: More straight forward questions like functions of the various parts of the microscope can be expected from this section

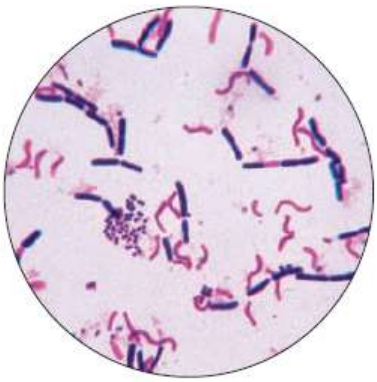
STAINING

Microorganisms often can be stained very satisfactorily by **simple staining**, in which a single staining agent is used or **Differential Staining**.

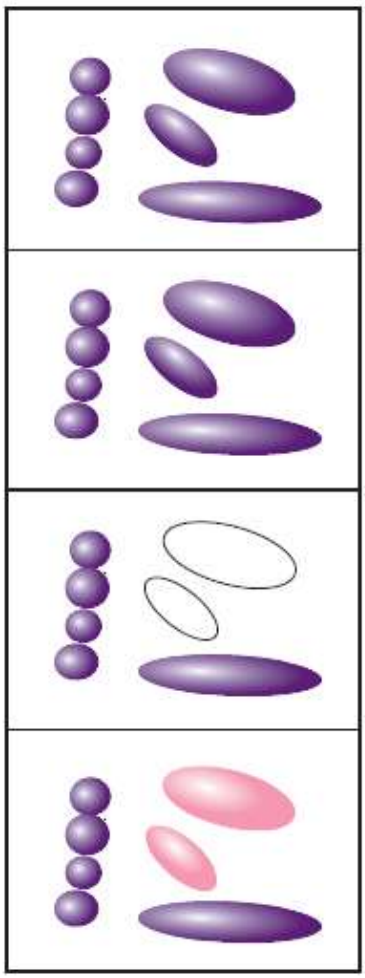
Differential staining procedures divide bacteria into separate groups based on staining properties.

Gram stain

- It was developed in 1884 by the Danish physician Christian Gram, is the most widely employed staining method in microbiology.
- It is **a differential** staining procedure because it divides bacteria into two classes—**gram negative and gram positive.**
- In the first step of the Gram-staining procedure, the smear is stained with the basic dye **crystal violet**, the primary stain.
- It is followed by treatment with an **iodine** solution functioning as a **mordant.** That is, the iodine increases the interaction between the cell and the dye so that the cell is stained more strongly.
- The smear is next decolorized by washing with **ethanol or acetone.**
- This step generates the differential aspect of the Gram stain; gram-positive bacteria retain the crystal violet, whereas gram negative bacteria lose their crystal violet and become colorless.
- Finally, the smear is counterstained with a simple, basic dye different in color from crystal violet.
- **Safranin**, the most common counterstain, colors gram-negative bacteria **pink or red** and leaves gram-positive bacteria **dark purple**



Gram stain. Purple cells are gram positive and pink cells are gram negative



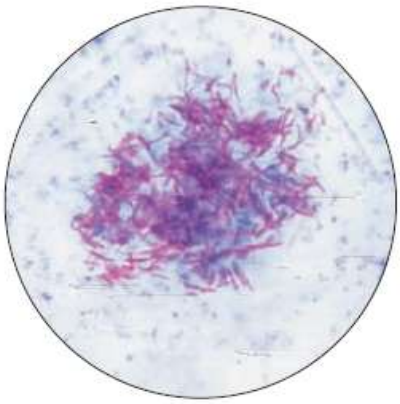
Steps in Staining	State of Bacteria
Step 1: Crystal violet (primary stain)	Cells stain purple.
Step 2: Iodine (mordant)	Cells remain purple.
Step 3: Alcohol (decolorizer)	Gram-positive cells remain purple; Gram-negative cells become colorless.
Step 4: Safranin (counterstain)	Gram-positive cells remain purple; Gram-negative cells appear red.

Gram Staining Procedure

Important notes: 1. The various chemicals used and the sequence in which they are used are very important. 2. Need to memorize the role of each chemical that is used in the staining procedure. 3. The Final coloration of the Gram Positive (Purple) and the Gram Negative (Pink) bacteria along with the examples of some common Gram positive and Gram-Negative bacteria are also very important

Acid-fast staining

- It is another important differential staining procedure.
- A few species, particularly those in the genus Mycobacterium do not bind simple stains readily and must be stained by heating with a mixture of **basic fuchsin** and **phenol** (the **Ziehl-Neelsen method**).
- Once basic fuchsin has penetrated with the aid of heat and phenol, acid-fast cells are not easily decolorized by an acid alcohol wash and hence remain red.
- This is due to the quite high lipid content of acid-fast cell walls.
- **Non-acid-fast bacteria** are decolorized by acid-alcohol and thus are stained blue by **methylene blue** counterstain.
- This method is used to identify *Mycobacterium tuberculosis* and *M. leprae*, the pathogens responsible for tuberculosis and leprosy, respectively.
- Negative staining, a technique that reveals the presence of the diffuse **capsules** surrounding many bacteria.
- Bacteria are mixed with **India ink or Nigrosin dye** and spread out in a thin film on a slide.
- After air-drying, bacteria appear as lighter bodies in the midst of a **blue-black** background because ink and dye particles cannot penetrate either the bacterial cell or its capsule.

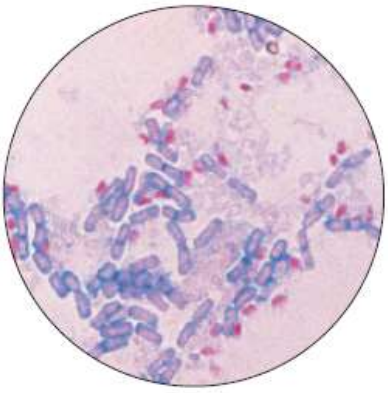


Acid-fast stain

Red cells are acid-fast and Blue cells are non-acid-fast.

Spore Staining

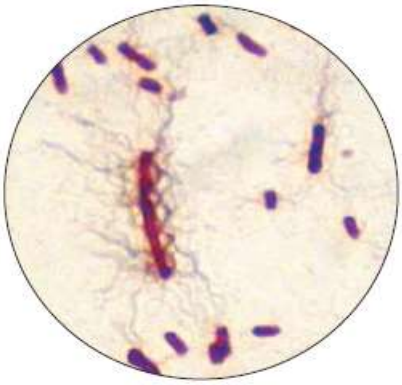
- In this method (**Schaeffer-Fulton procedure**), **endospores** are first stained by heating bacteria with **malachite green**, which is a very strong stain that can penetrate endospores.
- After **malachite green** treatment, the rest of the cell is washed free of dye with water and is counterstained with **safranin**.
- This technique yields a green endospore resting in a **pink or red** cell.



Endospore stain, showing endospores (red) and vegetative cells (blue)

Flagella Staining

- Flagella staining provides valuable information about the presence and **distribution pattern** of flagella on prokaryotic cells.
- Prokaryotic flagella are fine, threadlike organelles of locomotion that are about 10 to 30 nm in diameter
- This procedure employ coating flagella with mordants like **tannic acid** and **potassium alum**, and they are stained with **pararosaniline** (**Leifson method**) or basic fuchsin (**Gray method**).

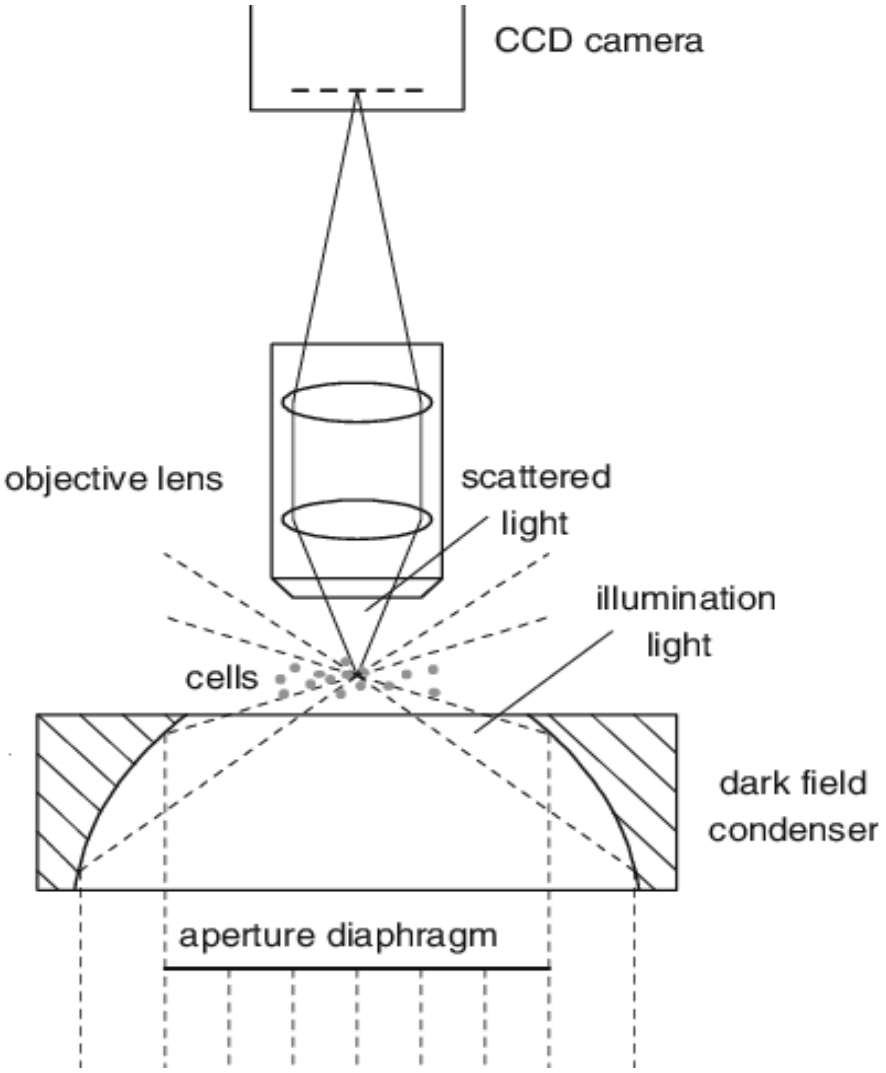


Flagellar stain of *Proteus vulgaris*. A basic stain was used to build up the flagella.

DARK FIELD MICROSCOPY

- The dark-field microscope allows a viewer to observe living, **unstained cells** and organisms
- It produces images of **brightly illuminated** objects on a black background
- Only light that has been reflected or refracted by the specimen forms an image
- A hollow cone of light is focused on the specimen in such a way that unreflected and unrefracted rays do not enter the objective.
- This technique has traditionally been used for viewing the outlines of objects in liquid media such as living spermatozoa, microorganisms, cells
- For lower magnifications, a simple darkfield setting on the condenser will be sufficient.

- For more critical darkfield imaging at a higher magnification, a **darkfield condenser** with a darkfield **objective lens** will be required.
- It also is used to identify certain bacteria like the thin and distinctively shaped *Treponema pallidum*, the causative agent of syphilis.
- If the object surface is of uneven thickness, hence different portions in an object absorb light to different extent then Phase contrast Microscopy will be used.

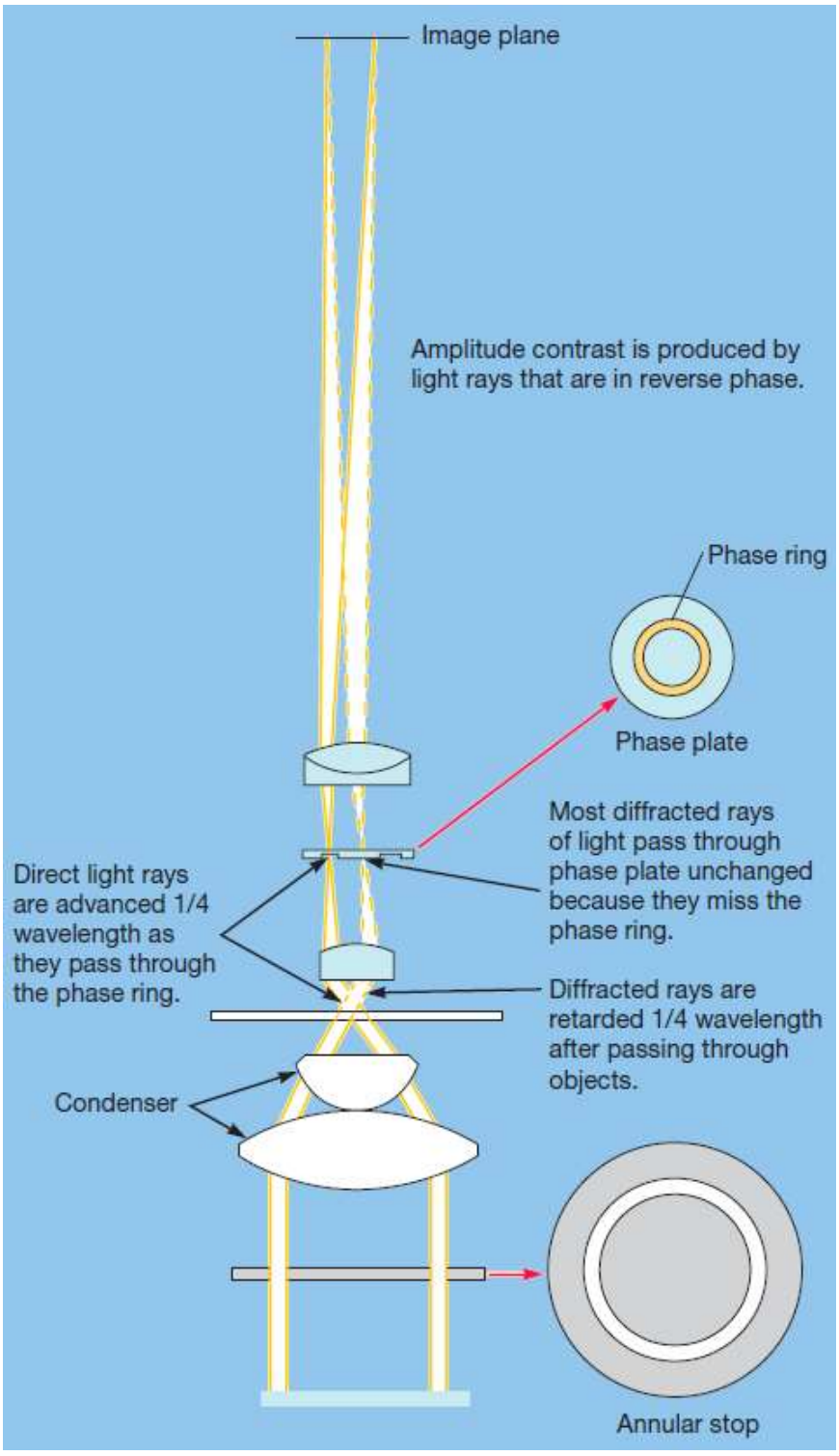


Dark Field microscope

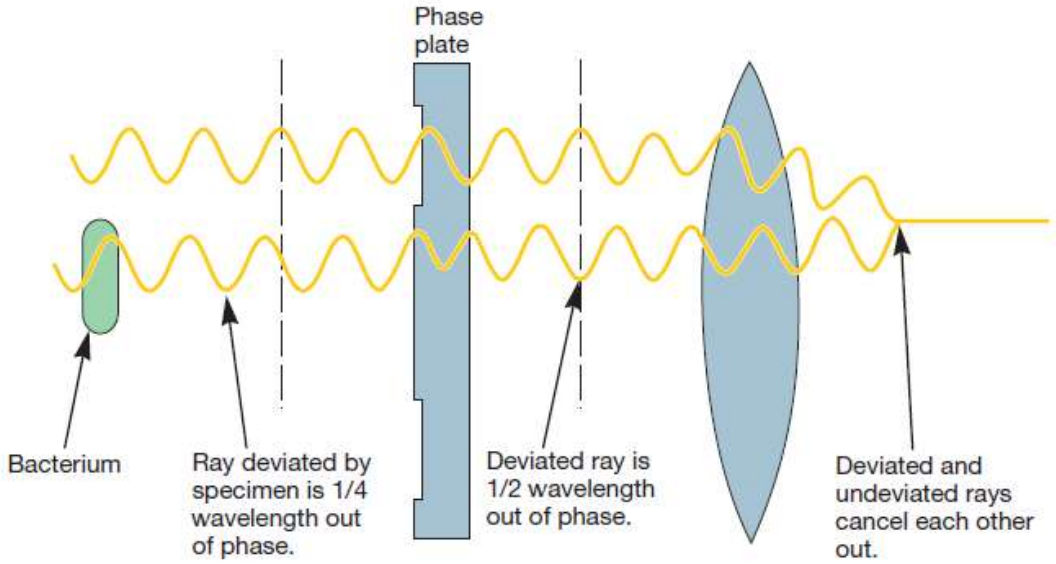
PHASE CONTRAST MICROSCOPY

- The visibility of the magnified specimen depends on contrast and resolution.
- variation in the light intensity (contrast) within an image occurs because different parts of the specimen absorb light to differing degrees.
- Brightness contrast arises from different degrees of absorption at different points in the specimen.
- Color contrast can also arise from absorption when the degree of absorption depends on the wavelength and varies from point-to-point in the specimen.
- For colorless specimens, as is the case for most biological material, contrast is achieved in various ways.
- The specimen may be stained, thus reducing the amplitude of certain light waves passing through the stained areas.
- However, this usually requires the fixation and staining of specimens.
- But fixation and staining kills the specimens. For colorless living specimens, contrast can be achieved in different ways.
- A living biological specimen is almost uniformly transparent, and therefore the intensity variation in the image will be poor.
- However, the light does not go through the specimen unaffected.
- The light going through the specimen will be shifted in phase due to scattering and diffraction. Our eyes cannot detect this phase shift.
- However, using phase contrast microscopy, we can develop a contrast based on the phase shift.
- Phase contrast is used for viewing unstained cells growing in tissue culture and for testing cell and organelle preparations for lysis
- The method images differences in the refractive index of cellular structures.
- Light that passes through thicker parts of the cell is held up relative to the light that passes through thinner parts of the cytoplasm.
- It requires a specialised phase condenser and phase objective lenses.

- Each phase setting of the condenser lens is matched with the phase setting of the objective lens.
- If object or specimen to be imaged has the ability to emit light, then it is observed with the help of a fluorescence microscope.



Phase Contrast Microscope

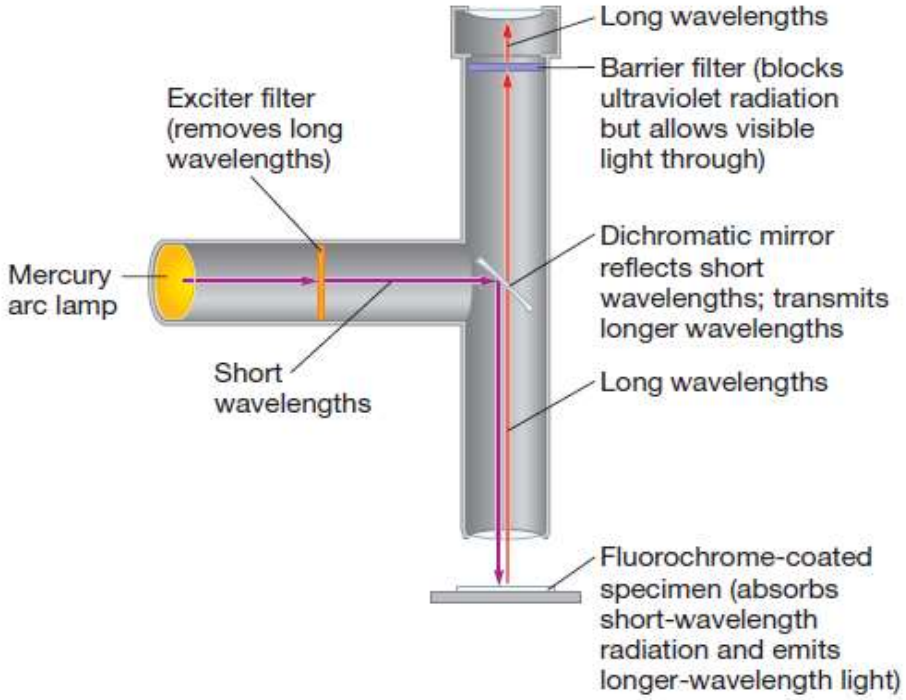


Production of contrast in Phase Contrast microscopy

FLUORESCENT MICROSCOPY

- Fluorescence microscopy is currently the most widely used contrast technique.
- The most commonly used fluorescence technique is called **epifluorescence light microscopy**, where 'epi' simply means 'from above'.
- Here, the light source comes from above the sample, and the objective lens acts as both condenser and objective lens.
- Fluorescence is popular because of the ability to achieve highly **specific labelling** of cellular compartments.
- The images usually consist of distinct regions of fluorescence (white) over large regions of no fluorescence (black), which gives excellent signal-to-noise ratios.
- The light source is usually a high pressure **mercury or xenon vapour lamp**, which emits from the ultraviolet into the red wavelengths.

- A specific wavelength of light is used to excite a fluorescent molecule or **fluorophore** in the specimen
- Light of longer wavelength from the excitation of the fluorophore is then imaged.
- This is achieved in the fluorescence microscope using combinations of **filters** that are specific for the excitation and emission characteristics of the fluorophore of interest.
- There are usually three main filters: an **exciter filter**, a **dichromatic mirror** (often called **a dichroic**) and a **barrier filter**, mounted in a single housing above the objective lens.
- The **exciter filter** transmits only the desired wavelength of excitation light.
- The **dichromatic mirror** reflects light of shorter wavelengths (i.e., the excitation light) but allows light of longer wavelengths to pass through.
- The **barrier filter** blocks ultraviolet radiation but allows visible light to pass through.
- For very tiny specimen (eg. viruses), the detailed internal structure can be observed only with electron microscopes.



Fluorescence microscope

ELECTRON MICROSCOPY

- The light microscopes are most important instrument for studying microorganisms.
- But even the best light microscopes have a resolution of only 0.2 μm
- Hence, they cannot be used for detailed studies of many microorganisms.
- Viruses, for example, are too small to be seen with light microscopes.
- Prokaryotes can be observed, but only their general shape and major morphological features are visible.

- The detailed internal structure of larger microorganisms also cannot be effectively studied by light microscopy.
- These overcome these limitations, Electron microscopes have been used that have much greater resolution.
- Electron Microscopes are scientific instruments that use a beam of highly energetic electrons to examine objects on a very fine scale.
- The electron microscope is able to achieve greater magnification and resolution because it uses a high voltage beam of electrons, whose wavelength is very much shorter than that of visible light.
- As a result, it is able to resolve points that are much closer together than is possible even with the very best light microscope.
- The resolving power of an electron microscope may be as low as 1–2 nm, enabling visualization of viruses and the internal structure of several cells.
- The greatly improved resolution means that specimens can be successfully magnified over 100000 times.
- Electron microscopes, generally use ring-shaped electromagnets as ‘lenses’ to focus the beam of electrons onto the specimen.
- Because the electrons would collide with, and be deflected by, molecules in the air, electron microscopes require a pump to maintain a vacuum in the column of the instrument.
- The examination can yield the following information:
- **Topography**

  The surface features of an object or "how it looks", its texture; direct relation between these features and materials properties (hardness, reflectivity etc.)
- **Morphology**

  The shape and size of the particles making up the object; direct relation between these structures and materials properties (ductility, strength, reactivity etc.)
- **Composition**

The elements and compounds that the object is composed of and the relative amounts of them; direct relationship between composition and materials properties (melting point, reactivity, hardness...etc.)

- **Crystallographic Information**

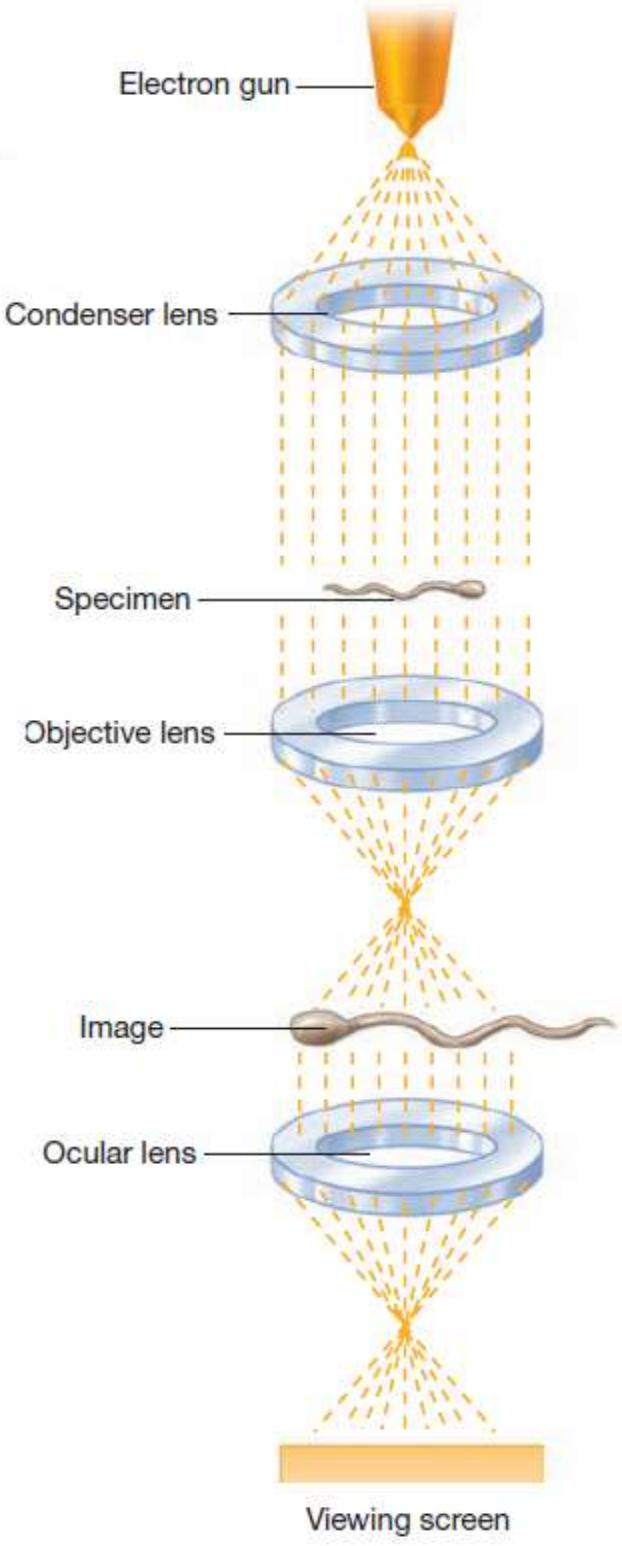
  How the atoms are arranged in the object; direct relation between these arrangements and materials properties (conductivity, electrical properties, strength etc.)

- There are two principal types of electron microscope, the transmission electron microscope (TEM) and the scanning electron microscope (SEM).

TRANSMISSION ELECTRON MICROSCOPY

- The transmission electron microscope (TEM) operates on the same basic principles as the light microscope but uses **electrons** instead of light.
- It consists largely of a tall, hollow cylindrical column through which the electron beam passes and a console containing a panel of dials that electronically control the operation in the column.
- The top of the column contains the **cathode**, a **tungsten wire filament** that is heated to provide a source of electrons.
- Electrons are drawn from the hot filament and **accelerated** as a fine beam by the high voltage applied between the cathode and anode.
- Air is pumped out of the column prior to operation, producing a vacuum through which the electrons travel.
- A beam of negatively charged electrons is focused by **electromagnetic lenses**, which are located in the wall of the column.
- The strength of the magnets is controlled by the current provided them, which is determined by the positions of the various dials of the console.

- The **condenser lenses** are placed between the electron source and the specimen, and they focus the electron beam on the specimen.
- The specimen is supported on a small, thin metal grid (3 mm diameter) that is inserted with **tweezers** into a grid holder, which is inserted into the column of the microscope.
- For examination of surface of microorganisms in detail, scanning electron microscope would be employed, that produce an image from electrons released from atoms on an object's surface.

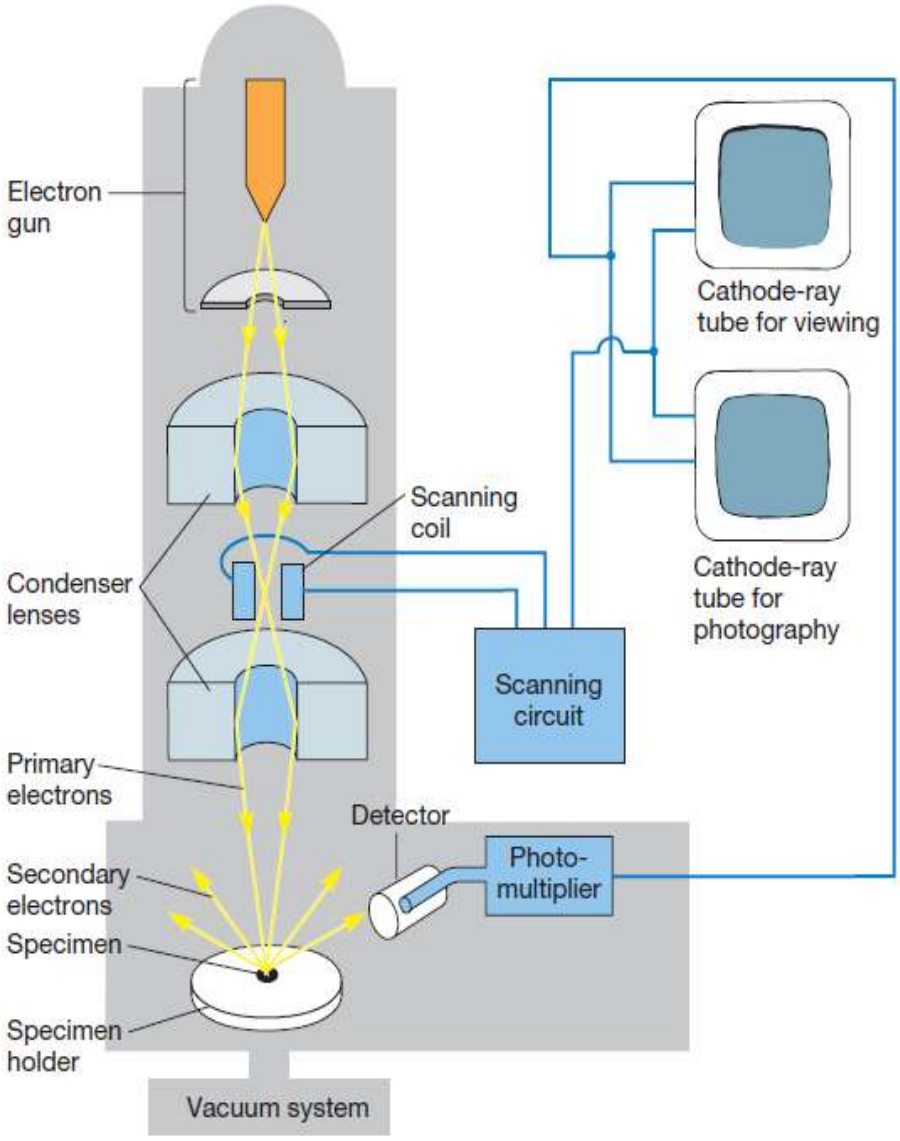


The transmission electron microscope (TEM)

SCANNING ELECTRON MICROSCOPY

- The scanning electron microscope (SEM) is utilized primarily to examine the surfaces of objects ranging in size from a virus to an animal head.
- The construction and operation of the SEM are very different from that of the TEM.
- Specimens to be examined in the SEM are fixed, passed through a series of alcohols, and then dried by a process of **critical-point drying.**
- At this point, there is no surface tension between the gas and the liquid.
- The solvent of the cells is replaced with a liquid transitional fluid (generally carbon dioxide), which is vaporized under pressure so that the cells are not exposed to any surface tension that might distort their three-dimensional configuration.
- Once the specimen is dried, it is coated with a thin layer of metal, which makes it suitable as a target for an electron beam.
- **Electrons** are **accelerated** as a fine beam that scans the specimen.
- The image is formed by electrons that are reflected back from the specimen (backscattered) or by secondary electrons given off by the specimen after being struck by the primary electron beam.
- These electrons strike a **detector** that is located near the surface of the specimen.
- Image formation in the SEM is **indirect.**
- In addition to the beam that scans the surface of the specimen, another electron beam synchronously scans the face of a **cathode-ray tube**, producing an image similar to that seen on a television screen.

- The electrons that bounce off the specimen and reach the detector control the strength of the beam in the cathode-ray tube.
- The more electrons collected from the specimen at a given spot, the stronger the signal to the tube and the greater the **intensity** of the beam on the screen at the corresponding spot.
- The result is an image on the screen that reflects the **surface topology** of the specimen because it is this topology that determines the number of electrons collected from the various parts of the surface.



The scanning electron microscope (SEM)



	Light microscope	Electron microscope
Main characteristics	Utilizing a microscope, optical lenses, and lights waves to enlarge small objects and producing resultant images.	Utilizing a microscope and electron beams to enlarge small objects and producing resultant images.
Advantages	- Cheap. - Small and portable. - Unaffected by magnetic fields. - Set-up is quick, simple, and only needs little expertise. - Rare distortion of materials during preparation. - Observing a material's natural color.	- Magnification of scanned objects over 500 000x. - Possibility of investigating greater field depth.
Disadvantages	- Magnification of scanned objects only up to 2000x. - Restricted field depth.	- Expensive. - Very large and therefore requires special operating rooms. - Affected by magnetic fields. - Setting up the materials requires expertise and is time-consuming. - Possibility of material distortion during preparation. - Produces black and white images.

Important notes: 1. The differences between the electron microscopes and the light microscopes in terms of their instrumentations and the advantages and disadvantages are important.